

ISM Note 8

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4/16/2026

1. Plasma Basics

First of all, we will study the plasma wave. Consider a charge neutral media, where

$$n_+ = n_- = n$$

If we perturb the plasma, the electrons and ions will be separated, and then there will be an excess of charge density, which is given by

$$\rho = en\xi$$

And the electric field can be derived by the Gauss's law, which is given by

$$\vec{E} = (4\pi en\xi)\hat{x}$$

And from the Newton's second law, we can thus get the equation of motion for electrons, which is given by

$$(m_e n \xi) \frac{d^2 \xi}{dt^2} = (ne\xi)(4\pi ne\xi)$$

This is a harmonics equation, and the solution is given by $\xi \propto e^{i\omega t}$. Here the frequency is called the **electron plasma frequency**, which is given by

$$\omega_p^2 = \frac{4\pi ne^2}{m_e}$$

Interestingly, this is also called **the cut-off frequency in plasma**. So what does this mean?

We can consider it this way: if you perturb the plasma with a frequency **lower** than the plasma frequency, the electron will rapidly bounce back and there will be no wave propagated in the plasma. Only if you perturb the plasma with a frequency **higher** than the

plasma frequency, the electrons are not capable to respond to the perturbation in time, in this case the wave can propagate.

There is another important thing in plasma, which is called **the Debye shielding**. Consider a test charge in the plasma, the test charge will attract opposite charges and repel like charges. Due to this interaction, there will be a cloud of opposite charges around the test charge, which can shield the electric field of the test charge. This means **the charges are mobile and can shield out external charge field**. Obviously, this shielding is not infinite, so the question is, what is the length scale of the neutral region of shielding?

This gives rise to **the Debye length**, which is given by

$$\lambda_D^2 = \frac{kT}{4\pi n e^2}$$

The Debye length is the characteristic length of the charge cloud. Outside the Debye sphere, we can treat the plasma as neutral as if the test charge does not exist. Therefore, a larger Debye length means the shielding is not efficient.

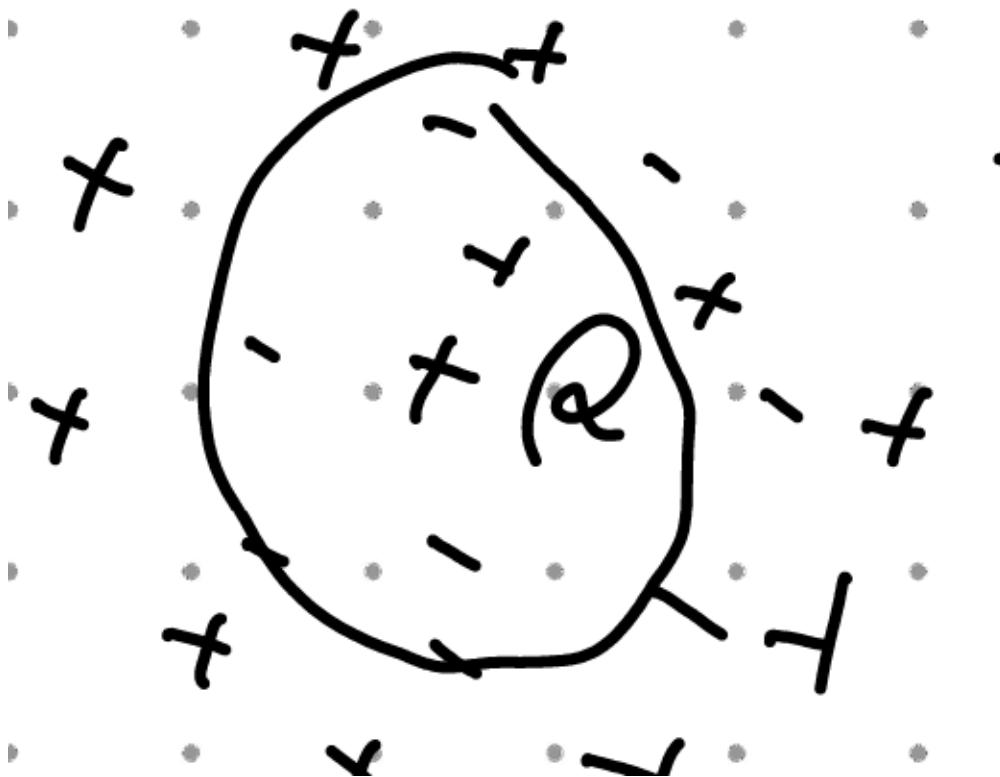


Fig 1. The Debye Shielding

The Debye length shows the competition between random motion and electromagnetic force. If the temperature is high, the random motion is strong, and the

Debye length is large, which means the shielding is not efficient. If the density is high, the electromagnetic force is strong, and the Debye length is small, which means the shielding is efficient.

Note that Debye shielding is valid in a statistical sense, which means on average the plasma is neutral, but locally there can be charge separation.

Single particle motion

We consider the gyro motion, which is the motion of a charged particle in a magnetic field. The equation of motion is given by

$$\frac{d\vec{p}}{dt} = q \frac{\vec{v}}{c} \times \vec{B}$$

If the particle is non-relativistic, $\vec{p} = m\vec{v}$, and the solution is given by

$$v_{x,y} = v_{\perp} e^{i\Omega t}$$

$$v_z = v_{\parallel}$$

where Ω is the **gyro frequency**, which is given by

$$\Omega = \frac{qB}{mc}$$

The **gyro radius** or Larmor radius is given by

$$r_g = \frac{\gamma v_{\perp}}{\Omega}$$

For a common astrophysical condition, the magnitude of some parameters are given by

$$B \sim \mu\text{G}$$

$$E_{CR} \sim \text{GeV}$$

$$r_g \sim \text{AU}$$

For example the gyro radius of the solar wind is about $\sim 10^3$ cm, while for ISM it is about $\sim 10^3$ cm, which is much smaller than the scale we care about, and for IGM it is about $\sim 10^7$ cm.

The gyro motion can be decomposed into two parts, the motion of guiding center plus the circular motion around the guiding center.

Particle Drifts

Let's impose an external electric field E on the plasma, then the equation of motion is given by

$$\frac{d\vec{p}}{dt} = q(\vec{E} + \frac{\vec{v}}{c} \times \vec{B})$$

Suppose the electric field is in the yz plane, and the magnetic field is in the z direction, then we can get the solution for the velocity, which is given by

$$\begin{cases} v_x = v_{\perp} e^{i\Omega t} - \frac{E_y}{B} \\ v_y = i v_{\perp} e^{i\Omega t} \\ v_z = v_{\parallel} + \frac{qE_z}{m} t \end{cases}$$

The first term is simply the **cyclotron motion**, and the important thing is the second term, which is called **the E cross B drift**. This means that if there exists an electric field, the particle drift will **be perpendicular to both the electric field and the magnetic field**. This is also called the side way drift, and it is given by

$$\vec{v}_E = c \frac{\vec{E} \times \vec{B}}{B^2}$$

This is independent of the charge and mass of the particle, so electrons and ions will drift together, thus **the plasma will still keep neutral and there's no net current density**.

Particle Drifts with Non-uniform Magnetic Field

Firstly, it is the **gradient B drift**. We know from classical electrodynamics class that the magnetic moment is given by $\vec{\mu} = \frac{1}{2} \frac{mv_{\perp}^2}{B}$, and the force is given by $\vec{F} = -\nabla(\vec{\mu} \cdot \vec{B})$. Therefore the drift velocity is given by

$$\vec{v}_{\nabla B} = \frac{v_{\perp}^2}{2\Omega} \frac{\vec{B} \times \nabla B}{B^2}$$

We can see that the drift velocity is related to **the perpendicular velocity of the particle**.

Another drift is the **curvature drift**. The curvature is defined as

$$\vec{\kappa} \sim \frac{\vec{R}}{R}$$

where $\vec{R}$ is the radius of the B field line.

And the curvature drift is given by

$$\vec{v}_c = -\frac{v_{\parallel}^2}{\Omega R} \frac{\vec{B} \times \vec{\kappa}}{B}$$

which means that if the magnetic field lines are curved, the particle will drift due to its inertia, and this is related to **the parallel velocity of the particle**. Essentially, it is caused by a kind of a centrifugal force.

2. Adiabatic Invariants

If the magnetic field is a constant or it changes slowly compared to the gyro motion, then there are some quantities that are conserved, which are called **adiabatic invariants**.

The first adiabatic invariant is the magnetic moment, which is given by

$$\mu = \frac{1}{2} \frac{mv_{\perp}^2}{B}$$

So why does this quantity is conserved? We can understand it from the Maxwell's equation. If the magnetic field changes slowly, we mean

$$\frac{1}{B} \frac{dB}{dt} \ll \Omega$$

Then we can get the electric field from Faraday's law, which is given by

$$\oint \vec{E} \cdot d\vec{l} = -\frac{1}{c} \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} = -\frac{1}{c} \frac{dB}{dt} \pi r_g^2$$

The left hand side is the work done by the electric field, and plugging this into Newton's second law, we can get the change of the kinetic energy, which is given by

$$\frac{d}{dt} \left(\frac{1}{2} m v_{\perp}^2 \right) = \frac{q\Omega}{2\pi} \frac{\pi r_g^2}{c} \frac{dB}{dt} = \mu \frac{dB}{dt}$$

Then the definition of the magnetic moment tells

$$\frac{d(\mu B)}{dt} = \frac{d}{dt} \left(\frac{1}{2} m v_{\perp}^2 \right)$$

Combining these two equations, we can get

$$\frac{d\mu}{dt} = 0$$

This is the so-called **first adiabatic invariant**.

Then we can have an important application of the first adiabatic invariant, which is **the magnetic mirror**. If the magnetic moment is a constant, then from definition we have

$$v_{\perp}^2 \propto B$$

Since the kinetic energy of the particle is also a constant, the parallel velocity of the particle in higher magnetic fields area will be smaller and smaller until zero, and since the Lorentz force still exist, the particle will be reflected back to the lower magnetic field region. This phenomenon is called **particle trapping**.

Usually not all the particles can be trapped in the magnetic mirror. To quantify this, we define a quantity

$$R_m = \frac{B_{max}}{B_{min}}$$

which is called the mirror ratio. The condition for the particle to be trapped is thus given by

$$R_m > \frac{1}{\sin^2 \theta}$$

where θ is the **pitch angle** of the particle, which is defined as $\tan \theta = \frac{v_{\perp}}{v_{\parallel}}$. So if the particle has a large pitch angle, it is easier to be trapped.

3. Magnetohydrodynamics (MHD)

If we want to describe the plasma on a large scale, we need to use the magnetohydrodynamics (MHD) and add the Lorentz force term into the momentum equation, which is given by

$$\rho \frac{d\vec{v}}{dt} = -\nabla P + \frac{1}{c} \vec{J} \times \vec{B} + \vec{F}$$

where $\vec{F}$ is external force, and the magnetic force term can be rewritten as

$$\frac{1}{c} \vec{J} \times \vec{B} = \frac{1}{4\pi} (\nabla \times \vec{B}) \times \vec{B} = -\nabla \left(\frac{B^2}{8\pi} \right) + \frac{1}{4\pi} (\vec{B} \cdot \nabla) \vec{B}$$

The first term is the **magnetic pressure**, which measures the magnetic energy carried by the magnetic field.

We can define a plasma beta, which is given by

$$\beta = \frac{P_g}{P_B}$$

This is the ratio of the gas pressure to the magnetic pressure, and it tells us the relative importance of the gas pressure and the magnetic pressure. **When we say a gas is "low beta", it means the magnetic field is dynamically important.**

The second term is the **magnetic tension**, which is the force **trying to straighten the magnetic field lines**. If we perturb the magnetic field lines, the magnetic tension will try to restore the magnetic field lines. This restoration will create a wave, which is called **the Alfvén wave**. The Alfvén wave is a transverse wave propagating along the magnetic field lines, and the velocity of the Alfvén wave is given by

$$v_A = \frac{B}{\sqrt{4\pi\rho}}$$

This tells us **how fast the information can propagate along the magnetic field lines**.

Like the case in hydrodynamics, we need another equation to tell how the magnetic field evolves. This is given by **the induction equation**.

Firstly, the Ohm's law is given by

$$\vec{j} = \sigma(\vec{E} + \frac{1}{c}\vec{v} \times \vec{B})$$

where $\sigma \sim \frac{ne^2}{m}\tau_{\text{coll}}$ is the conductivity, and $\tau_{\text{coll}} \sim \frac{\lambda}{v}$ is the collision timescale.

Then we plug Ohm's law into Faraday's law, and using Ampere's law, we can get the induction equation, which is given by

$$\frac{\partial \vec{B}}{\partial t} = \nabla \times (\vec{v} \times \vec{B}) + \eta \nabla^2 \vec{B}$$

where $\eta = \frac{c^2}{4\pi\sigma}$ is the **magnetic diffusivity** or magnetic resistivity.

We need to give a remark on Ampere's law. We assume the displacement current term $\frac{\partial \vec{D}}{\partial t}$ is negligible, and for good reason.

One important reason is **the plasma is a good conductor**. According to Ohm's law, the electric field $\vec{E}$ must be very small to avoid a large current.

Another reason is **the plasma is non-relativistic**. Therefore, the electric field is smaller than the magnetic field by a factor of order v/c , so it is negligible.

One important implication of the induction equation is the so-called **flux freezing**. The magnetic flux is given by

$$\Phi_B = \int \vec{B} \cdot d\vec{S}$$

Exerting a time derivative on both sides, we can get the rate of change of the magnetic flux

$$\frac{d\Phi_B}{dt} = \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} + \int \vec{B} \cdot \frac{\partial}{\partial t}(d\vec{S})$$

And since

$$\frac{\partial}{\partial t}(d\vec{S}) = (\vec{v} \cdot \nabla)(d\vec{S}) - d\vec{S}(\vec{v} \cdot \nabla)$$

The first term is the **stretching** of the surface, and the second term is the **compression** of the surface. Then we can plug this equation into the second term of the rate of change of the magnetic flux, and we can get

$$\begin{aligned} \int \vec{B} \cdot \frac{\partial}{\partial t}(d\vec{S}) &= \int \vec{B} \cdot \nabla \times (\vec{v} \times \vec{S}) \\ &= \int \nabla \times (\vec{B} \times \vec{v}) \cdot d\vec{S} \\ &= \oint (\vec{B} \times \vec{v}) \cdot d\vec{l} \end{aligned}$$

Then we plug the induction equation into the rate of change of the magnetic flux, and can get

$$\frac{d\Phi_B}{dt} = \int \eta \nabla^2 \vec{B} \cdot d\vec{S}$$

When $\eta \rightarrow 0$, the magnetic flux is conserved, which means **the magnetic field lines through some loops are tied to plasma and not slipping off it**. This is the so-called flux freezing. Usually when this condition is satisfied, we say it is ideal MHD.



Fig 2. The flux freezing.

Resistive Limit

Note that the evolution equation of the magnetic field has the same form as the standard diffusion equation, which is given by

$$\frac{\partial \vec{B}}{\partial t} = \eta \nabla^2 \vec{B}$$

So if $\eta \neq 0$, the magnetic field can diffuse and slip off from the loop due to diffusivity, and in this case the flux freezing will not hold.

One direct result from this is the **magnetic reconnection**, which is the process where the magnetic field lines can break and reconnect, converting the magnetic energy into thermal energy. This is an important source for heating in the ISM. The magnetic reconnection occurs commonly near the surface of the star and heat the gas.